GRADUATION PROJECT PROPOSAL

Basmati · بصمتي

Daily carbon footprint for people and a government facility: a phone app and an IoT energy monitor, piloted on the Computer Science department floor



TEAM	SUPERVISOR	PILOT SITE	DATE
[9 team members]	[Supervisor name]	CS department floor	October 2026

THE PROBLEM

Nobody sees the footprint of a person or of a public building

No idea of size or source

Neither people nor facility managers know how much CO_2e they produce, or which habit or room produces most of it.

One bill, too late

A government building gets one monthly bill for the whole building: no floor, no hour, no person, and it arrives after the energy is used.

Not built for Egypt

Personal calculators are yearly questionnaires with foreign factors: no microbus, no koshary, no Egyptian grid.

Goal: measure the CS floor and each member's day automatically, with Egyptian factors and real data, and show where the waste is.

CARBON FOOTPRINT

Egypt is below the world average, but above the 2030 target

A **carbon footprint** is the total greenhouse gas caused by a person, product or building over a period, measured in kg of CO_2-equivalent (CO_2e).

Every activity converts to CO_2e through an **emission factor**: kWh of electricity, km travelled, kg of food.

DAILY BUDGET PER PERSON

2,500 kg ÷ 365 ≈ 6.8 kg CO_2e

3.4 t per person per year in Egypt (2023)

6.6 t world average per person (2023)

2.5 t 1.5 °C lifestyle target per person by 2030

Sources: EDGAR 2024 report [27]; Hot or Cool Institute, 1.5-Degree Lifestyles [26]

OUR SOLUTION

One app and one floor monitor, two daily numbers

Transport

Commute detected from GPS and motion sensors; one tap to confirm when unsure

Food

Photo of the plate, at the cafeteria or at home; the model names the dish

Floor electricity

ESP32 monitors on the floor's electrical board; each person gets a share for their hours there

Facility dashboard

Daily kWh, CO_2e, after-hours waste and a forecast for the department head

Every night: each person's day against a 6.8 kg budget in the app, and the floor's day against its own baseline on a display at the floor entrance, in green, yellow or red.

SYSTEM ARCHITECTURE

The phone and the floor monitors feed one server

INPUTS

Trip tracking
GPS + transport model on phone

Food camera
meal model on phone (LiteRT)

Floor check-in
geofence: hours on the floor

Floor monitors
2 × ESP32 + CT clamps on the board

SERVER · FASTAPI

Trip and meal API
REST, from the phone

Energy ingest
MQTT · Mosquitto

Daily footprint engine
personal and floor totals

Forecast and waste
LightGBM + after-hours rules

Database
PostgreSQL + TimescaleDB

OUTPUTS

Personal summary
total, items, color in the app

Entrance display
LED ring: the floor's day

Facility dashboard
for the department head

Waste alerts
to the floor technician

WORKFLOW

The app senses, the user confirms, the server adds up



When	What happens
A trip ends	Mode classified on the phone; one-tap question if confidence is below 70%
A meal photo is taken	Top 3 dishes shown; the user picks the dish and the size
Arriving at the department	Geofence check-in counts the hours on the floor (no location is kept)
Every 10 seconds	Floor monitors send current per phase and per circuit
After working hours	Load above the base level outside the timetable raises a waste alert
11 pm	Personal total and color in the app; the floor’s total and color on the entrance display
Every week	A personal tip; a floor report to the department head

DATA FLOW

Three pipelines, two daily numbers

Transport

GPS and motion sensors → Trip segments → Mode classifier → Trip record

GPS points and sensor windows → trips → mode and km on the phone → record sent by REST (raw GPS stays on the phone)

Food

Plate photo → MobileNet on phone → Top 3 + size → Meal record

photo → 224 × 224 image → dish probabilities → user picks dish and portion → record (the photo stays on the phone)

Energy

CT clamps on the floor board → MQTT every 10 s → TimescaleDB → kWh, forecast, waste

current samples → Irms per phase, W, Wh → JSON message → time-series table → daily kWh, forecast and after-hours waste

Daily footprint engine

+ hours on the floor, timetable, home bill

App, entrance display, dashboard, alerts

DATA FLOW · RECORDS

What travels between the components

Record	Key fields	Rate	Stored in
Meter reading	device, circuit, time, Irms, W, Wh	every 10 s	TimescaleDB
Trip record	user, start, end, km, mode, confidence, passengers	per trip	PostgreSQL
Meal log	user, dish, portion, time	per meal	PostgreSQL
Floor presence	user, date, hours on the floor	daily	PostgreSQL
Timetable	room, day, hour, expected people	per term	PostgreSQL
Waste event	circuit, start, end, kWh, confirmed	when found	PostgreSQL
Daily footprint	user or floor, date, items, total, color	daily	PostgreSQL

Privacy by design: photos and locations never leave the phone, presence is stored as hours only, volunteers are IDs instead of names, and raw GPS is deleted once the trip is summarized.

EQUATION 1 · THE CORE RULE

Every footprint is activity × emission factor

$$\mathrm{CO_2e} = \sum_i A_i \times \mathrm{EF}_i$$

A_i is the amount of activity i (kWh, km, kg) and EF_i its emission factor in kg CO_2e per unit.

PERSONAL BUDGET = 2,500 kg ÷ 365 ≈ 6.8 kg

Green	Yellow	Red
≤ 4.8 kg (70%)	≤ 6.8 kg	> 6.8 kg

Activity	Unit	kg CO_2e / unit
Egypt grid electricity (2024)	kWh	0.45
Petrol car, ÷ passengers	km	0.17
Microbus or bus	passenger-km	0.10
Metro	passenger-km	0.06
Walking or cycling	km	0
LPG cylinder (12.5 kg)	cylinder	37.5
Beef (dairy to beef herd)	kg	33 to 99.5
Chicken	kg	9.9
Rice	kg	4.5
Lentils and beans	kg	1.8

EQUATION 2 · A PERSON'S DAY

One person's daily footprint D, in kg CO_2e

$$D = \sum_{\text{trips}} d \cdot f_{\text{mode}} + \sum_{\text{meals}} f_{\text{dish}} \cdot s$$
$$+0.45 \cdot E_{\text{floor}} \cdot \frac{h_i}{H} + \frac{0.45E_{\text{home}} + G}{N_{\text{home}}}$$

d trip distance (km); **f_{mode}** mode factor

E_{floor} floor kWh per day from the monitors

E_{home} home kWh per day from the monthly bill (optional)

f_{dish} meal footprint; **s** portion 0.7, 1 or 1.4

h_i your hours on the floor; **H** all person-hours

G home gas: cylinders × 37.5 ÷ 30; **N_{home}** members

Transport	Food	Floor share	Home	Example day total
2.0	2.6	0.6	1.2	6.4 kg · 94% · yellow

EQUATION 3 · MEALS

A meal’s footprint comes from its ingredients

$$f_{\text{dish}} = \sum_{j} w_j \cdot \text{EF}_j \qquad F_{\text{meal}} = f_{\text{dish}} \cdot s$$

w j = raw weight of ingredient j in kg · EF j = its factor (Poore and Nemecek) · s = portion size chosen by the user

Dish (medium plate)	Ingredients, raw weight	kg CO_2e
Koshary	rice 100 g, lentils 50, pasta 50, tomato 50, onion 30	0.73
Chicken with rice	chicken 150 g, rice 100	1.93
Beef with rice	beef 150 g, rice 100	5.4 to 15.4

The team builds this table for 20 dishes, including the faculty cafeteria menu. Beef uses the dairy-herd factor (33) as a conservative value.

EQUATION 4 · THE FLOOR

The floor’s footprint, intensity and after-hours waste

$$C_{\text{floor}} = 0.45 \cdot E_{\text{floor}} \qquad I = \frac{C_{\text{floor}}}{H} \qquad r = \frac{E_{\text{floor}}}{E_{\text{base}}}$$

$$W = \sum_{t \notin T_{\text{work}}} \max\left(0, P_t - P_{\text{base}}\right) \cdot \Delta t$$

E_{floor} floor kWh per day, all three phases

I kg CO_2e per person-hour; **H** person-hours from the timetable

E_{base} same weekday, first 3 weeks (before feedback)

T_{work} timetable hours; **P_{base}** night base load (servers, fridges)

Green	Yellow	Red
r ≤ 0.9	r ≤ 1.0	r > 1.0

Example: 330 kWh → 149 kg CO_2e, 0.10 kg per person-hour; 40 kWh after hours = 18 kg wasted.

EQUATION 5 · ENERGY MONITOR

From current samples to calibrated kWh on three phases

$$I_{\text{rms}} = \left(\frac{1}{N} \sum_{n=1}^{N} i_n^2 \right)^{1/2}$$

$$P = k \cdot V \cdot \sum_{p=1}^{3} I_{\text{rms},p} \qquad E = \sum_{t} P \cdot \Delta t \qquad k = \frac{E_{\text{ref}}}{E_{\text{dev}}}$$

1. Monitor A: a clamp on each of the floor's three main phases.
2. Monitor B: clamps on lab ACs, lab PCs and lighting.
3. Power uses V = 220 V and k, updated weekly from a reference reading.
4. Target: weekly error below 10%.

AI MODELS

Four core models and one optional

Model	Input	Method	Runs on	Priority
Transport mode	Speed, stops, acceleration, sensor vibration, GPS loss	Activity Recognition API, then Random Forest or LightGBM	Phone	**Core**
Meal recognition	Plate photo	Transfer learning on MobileNetV3 or EfficientNet-Lite0, LiteRT int8	Phone	**Core**
Floor forecast	Hourly kWh, hour, weekday, temperature, timetable	LightGBM; LSTM if there is enough data	Server	**Core**
After-hours waste	Load per circuit, timetable, forecast	Rules first, then Isolation Forest on forecast errors	Server	**Core**
Appliance breakdown (NILM)	Floor main power	Disaggregation, checked against monitor B	Server	**Optional**

Rule: every model must beat a simple baseline measured first, on people and days not used in training.

DATASETS · PUBLIC

Public data for pretraining and benchmarks

Dataset	Year	Contents	Our use
ImageNet	2012	About 1.28M training images, 1,000 classes	Pretrained MobileNetV3 weights
Food-101	2014	101,000 images of 101 dishes	Food pretraining and benchmark
SHL	2018	Phone sensor data, 8 transport and locomotion modes	Pretrain the transport classifier
GeoLife	2012	GPS trajectories from 182 users, part with mode labels	Benchmark trip features
Building Data Genome 2	2020	Hourly meters of 1,636 non-residential buildings, 2 years	Pretrain and benchmark the forecast
UK-DALE	2015	Whole-house and appliance demand, 5 UK homes	NILM benchmark (optional)
Poore and Nemecek	2018	Global food life-cycle emission factors	Dish footprints
OpenStreetMap	—	Locations of Cairo metro stations	Metro geofences
Open-Meteo	—	Hourly temperature at the faculty	Forecast input

DATASETS · COLLECTED BY THE TEAM

Egyptian data, collected with consent and approval

Data	Amount	How	Weeks
Floor electricity	6 channels every 10 s; about 2,350 hourly readings each	Monitors on the floor board; weekly reference reading	6–20
Timetable and presence	Room schedule; hours on the floor per volunteer	Department administration; app check-in	2–20
Building bills	12–24 months of the building's electricity bills	Official request to the faculty administration	2–4
Food photos	200+ per dish: 3,000 photos from 30 people	Volunteers photograph meals and pick the dish name	4–10
Labeled trips	20 volunteers × 2 weeks ≈ 560 trips; 80+ per mode	Volunteer mode records GPS and sensors; user picks the mode	5–9
Waste events	About 30 labeled after-hours events	Staged with the technician, plus confirmed alerts	8–18

Split 70 / 15 / 15 by person and by week, never by photo or hour, so tests reflect new people and new days.

Consent forms, IDs instead of names, raw GPS deleted after feature extraction, right to withdraw.

PERFORMANCE METRICS

How each kind of model is measured

Classification

transport mode per trip; waste events found or missed

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \qquad \text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \qquad F_1 = \frac{2PR}{P+R}$$

Regression

floor forecast, monitor error, trip distance

$$\text{MAE} = \frac{1}{n}\sum |\, y_i - \hat{y}_i \,| \qquad \text{RMSE} = \left(\frac{1}{n}\sum (y_i - \hat{y}_i)^2\right)^{1/2} \qquad \text{MAPE} = \frac{100\%}{n}\sum \frac{|y_i - \hat{y}_i|}{y_i}$$

Top-k accuracy

meal recognition: is the true dish in the model's k best guesses?

$$\text{Top-}k = \frac{1}{n}\sum_{i=1}^{n} 1[y_i \in \text{top}_k(x_i)]$$

PERFORMANCE TARGETS

What “working” means for each part

Part	Metric	Target
Transport mode	Accuracy per trip	**At least 85%**
Trip distance	Error vs. true distance	**Below 10%**
Meal recognition	Top-1 / Top-3 accuracy	**At least 80% / 95%**
Meal speed	Time on a mid-range phone	**Under 1 second**
Floor monitor	Weekly error vs. reference reading	**Below 10%**
Floor forecast	Daily MAPE	**Below 15%, and beats last week**
Waste detection	Recall / precision on labeled events	**At least 90% / 80%**
Battery	App use per day	**Below 5%**
Real data	Volunteers and site	**20 volunteers, 1 floor, 14 weeks**

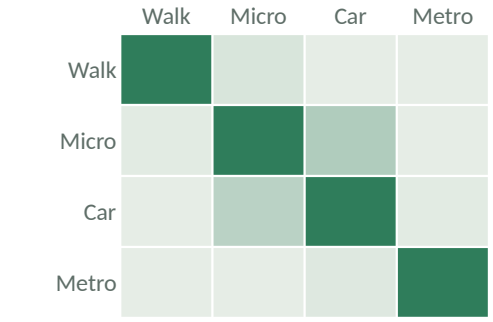
CURVES

The four curves we will report



Training vs. validation loss

Meal model, by epoch. A growing gap between the lines means overfitting.



rows: true mode · columns: predicted

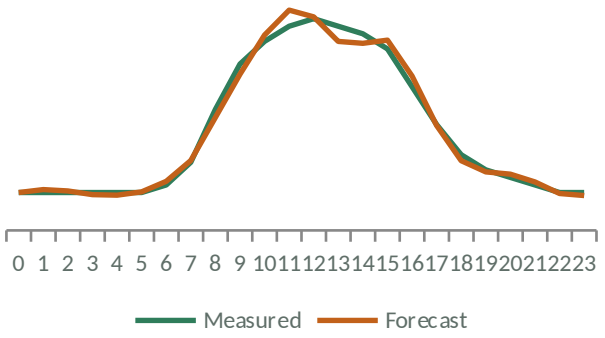
Confusion matrix

Transport modes: walk, microbus, car, metro. Shows which modes get mixed up.



Monitor vs. reference

Weekly floor kWh from our monitor beside the reference reading. The gap is the monitor error.



Predicted vs. actual kWh

One working day on the floor, hour by hour. The gap is what MAPE measures.

Illustrative shapes only, not results. The real curves come from our test data in weeks 10 and 18.

MEASURING AN EDUCATIONAL INSTITUTION

Universities and schools measure their footprint in three scopes

SCOPE 1 · DIRECT

Fuel burned on site

Gas for heating and kitchens, diesel generators, the institution’s buses, refrigerant leaks

Basmati: Optional manual entry

SCOPE 2 · ELECTRICITY

Purchased electricity

Labs, PCs, air conditioning and lighting

Basmati: Measured every 10 s by the floor monitors

SCOPE 3 · VALUE CHAIN

Everything else

Student and staff commuting, food, purchases, waste, water, business travel

Basmati: Commuting and meals per person, from the app

Standards: GHG Protocol Corporate and Scope 3 Standards; ISO 14064-1:2018; ISO/TR 14069. Emissions = activity × emission factor, then normalized per person or per m^2.

How to measure the footprint of a university

Paper	Year	What we take from it
[1] Valls-Val and Bovea — Carbon footprint in higher education institutions: a literature review	2021	Scopes, methods and sources used by universities
[3] Helmers, Chang and Dauwels — Carbon footprinting of universities worldwide, Part I	2021	20 universities; standardized metrics needed to compare
[4] Valls-Val and Bovea — CO2UNV: a carbon footprint tool for universities	2022	Scopes 1–3 per university or per building
[6] Herth and Blok — Universities' direct and indirect emissions (TU Delft)	2023	106 kt, 80% Scope 3; spending-based estimates are weak
[7] Kiehle et al. — Carbon footprint of the University of Oulu	2023	Hybrid input-output and LCA; heating is the largest item
[9] Abdelbasier et al. — Monitoring and mitigation in higher education (Ain Shams Univ. journal)	2022	Egyptian review; electricity is a major share

Numbers in brackets match the references at the end of the deck, with DOIs and links.

Measured campuses: commuting and buildings dominate

Institution	Country	Year	t CO_2e	Per person	Largest source	Ref.
Clemson University	USA	2014–17	≈ 95,400	—	Electricity and Scope 3, ≈ 40% each	[11]
Univ. Nacional de Colombia	Colombia	2019	7,251	0.43 t	Transport 58.5%; Scope 3 83%	[14]
NED University	Pakistan	2017	≈ 21,500	1.79 t	Scope 3 85.6%	[13]
TU Delft	Netherlands	2018	≈ 106,000	—	Scope 3 ≈ 80% (purchases)	[6]
University of Oulu	Finland	2019	19,072	—	District heating	[7]
University of Bologna	Italy	2020	16,467	—	IT purchases ≈ 50% (transport 74% in 2018)	[16]
TU Cluj-Napoca	Romania	2023	7,229	0.36 t	Natural gas for heating	[18]
Sichuan Univ. Jinjiang College	China	2023	44,584	1.89 t	Electricity	[19]
Port Said University, Faculty of Science	Egypt	2023–24	—	—	Scope 3 (commuting) ≈ 97%	[22]

Boundaries and years differ, so totals are not directly comparable; per-person figures are the fairer comparison.

Schools, Egypt and continuous measurement

Paper	Year	What we take from it
[20] Wagner et al. — Student-led carbon footprint assessment in schools (Schools4Future, Germany)	2021	Students measure their school: mobility, heating, electricity, canteen
[21] Adhikari and Pokhrel — Carbon footprint of schools in Pokhara, Nepal	2025	8 schools, GHG Protocol; school buses are the largest item
[22] Badr, El Beheary and Hassan — Carbon footprint of Egyptian universities: Port Said University	2026	Egyptian government university, per faculty; commuting ≈ 97%
[23] Safwat et al. — Energy data for higher-education buildings in Egypt	2024	Monthly data per building often missing
[24] Hamada, Hong and Raslan — Cairo's public administrative buildings	2025	About 40% cooling, 60% lighting and equipment
[25] García-Monge et al. — IoT monitoring on a smart campus in Spain	2023	Continuous measurement; 40–70% HVAC savings possible

Numbers in brackets match the references at the end of the deck, with DOIs and links.

Personal footprint apps, transport and food

Paper	Year	What we take from it
[28] Ajufo and Bekaroo — TCTracker: an automated personal carbon footprint calculator for transport	2021	GPS + AI app; tested with 40 users
[29] Bekaroo et al. — Calculating carbon emissions from personal travelling: key calculators	2021	10 calculators: inconsistent and opaque
[30] Zhang et al. — SpeedTransformer for transport mode, tested in the CarbonClever app	2026	348 users tracked for personal footprints
[31] Sadeghian, Håkansson and Zhao — Transport mode detection from GPS: a review	2021	Labeled trips are the bottleneck
[33] Wang and Jiang — T2Trans: temporal convolutional network on phone sensors	2022	Macro F1 86.42% on SHL, 8 modes
[34] Tahir and Loo — Survey of image-based food recognition and volume estimation	2021	Deep CNN features; mobile food logging

Numbers in brackets match the references at the end of the deck, with DOIs and links.

Energy monitoring, forecasting and waste detection

Paper	Year	What we take from it
[36] Ismail and Sahari — Real-time energy monitoring with ESP32 + SCT-013	2024	Worst-case error ≈ 12.4%, so we calibrate weekly
[37] Miller et al. — ASHRAE Great Energy Predictor III: overview and results	2020	Gradient boosting (LightGBM) wins
[38] Anand et al. — Occupancy-based energy modelling for institutional buildings	2021	Occupancy matters; gradient boosting is fast
[39] Himeur et al. — AI-based anomaly detection of energy use in buildings: a review	2021	Methods for finding waste
[40] Azad, Rajabi and Estebsari — NILM with deep neural networks: a review	2023	Optional appliance breakdown

Numbers in brackets match the references at the end of the deck, with DOIs and links.

RESEARCH GAP

Campuses measure their footprint once a year from bills and surveys; nothing measures a public building and its people every day.



Daily, not yearly

Scope 2 measured every 10 s and commuting measured per trip, instead of one yearly inventory and a commuting survey.



Local factors and data

Microbus, metro, Egyptian dishes and the Egyptian grid, with data we collect ourselves.



Per person and per floor

Each person’s real share of the floor, plus after-hours waste alerts for the department head.

TIMELINE · 20 WEEKS

A full day computed automatically by week 10

Week 2

Interfaces and floor survey

MQTT, API, model inputs; rooms, labs, ACs and the board

Week 6

Monitors and volunteers

2 monitors on the floor board; volunteer app with 20 people

Week 10

First full version

A whole day computed end to end with the first models

Week 16

Feature freeze

No new features; fixing and measuring only

Week 20

Final delivery

Final numbers, demo and report draft

Weeks 1–10 · build

- Interfaces, screens and database
- Floor survey, official approval, installation in week 6
- Trip tracking, food camera, daily screen
- Volunteer mode and data collection

Weeks 13–20 · improve and measure

- Improve the models; run them on the phone
- Floor forecast, waste alerts and dashboard
- Battery tests and final evaluation
- Report, demo and defense

Weeks 11–12 are kept light for mid-year exams.

TEAM AND BUDGET

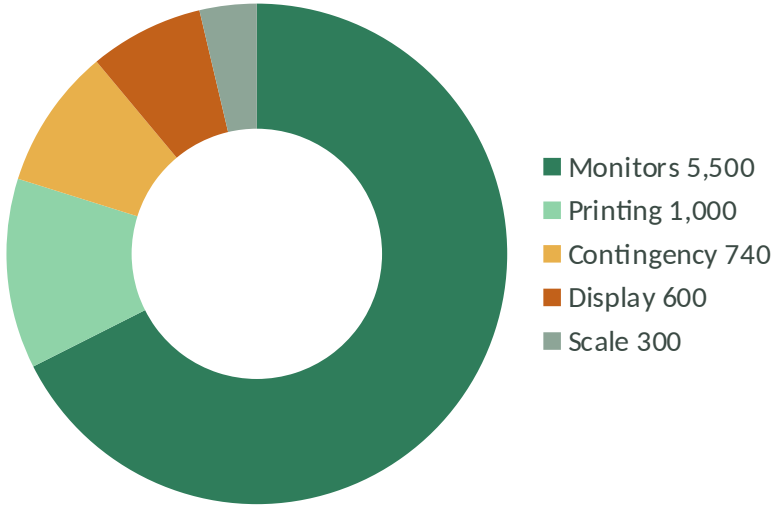
Nine students, four teams, about 8,100 EGP

Team	People	Owns
Mobile app	3	Trip tracking; food camera and on-phone models; screens, check-in and volunteer mode
AI	2	Meal model and dish database; transport model; waste detection
Backend	2	API, database, daily calculation, MQTT, forecast, dashboard
Hardware	2	Circuits and enclosures; installation with the faculty electrician; calibration; entrance display

Server on a faculty server or a free cloud tier (0 EGP). Prices are estimates at the time of writing.

ESTIMATED BUDGET

8,140 EGP



NEXT STEPS

What we need to start week 1

Decisions we need from you

- Approval of the idea and the pilot on the CS floor
- An official letter to the faculty administration for installation
- Access to the timetable and past electricity bills
- Consent form for 20 volunteers
- Final team names and roles

Our next two weeks

- Interfaces: MQTT, API and model inputs
- Floor survey: rooms, labs, ACs, PCs and the board
- Order parts; test one clamp on a kettle in the lab
- Dish list with the cafeteria menu; consent form

Measuring universities: reviews, methods and tools

[1] Valls-Val, K., & Bovea, M. D. (2021). Carbon footprint in higher education institutions: a literature review and prospects for future research. Clean Technologies and Environmental Policy, 23, 2523–2542. doi:10.1007/s10098-021-02180-2

[2] Li, Z., Chen, Z., Yang, N., Wei, K., Ling, Z., Liu, Q., Chen, G., & Ye, B. H. (2021). Trends in research on the carbon footprint of higher education: a bibliometric analysis (2010–2019). Journal of Cleaner Production, 289, 125642. doi:10.1016/j.jclepro.2020.125642

[3] Helmers, E., Chang, C. C., & Dauwels, J. (2021). Carbon footprinting of universities worldwide: Part I—objective comparison by standardized metrics. Environmental Sciences Europe, 33, 30. doi:10.1186/s12302-021-00454-6

[4] Valls-Val, K., & Bovea, M. D. (2022). Carbon footprint assessment tool for universities: CO2UNV. Sustainable Production and Consumption, 29, 791–804. doi:10.1016/j.spc.2021.11.020

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[6] Herth, A., & Blok, K. (2023). Quantifying universities' direct and indirect carbon emissions: the case of Delft University of Technology. International Journal of Sustainability in Higher Education, 24(9). doi:10.1108/IJSHE-04-2022-0121

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[8] Deda, D., Gervásio, H., & Quina, M. J. (2025). Advancing carbon footprint in higher education: an integrated assessment model. The International Journal of Life Cycle Assessment. doi:10.1007/s11367-025-02514-y

Methods and measured universities

[9] Abdelbasier, A. M., Kamel, M. M., Donia, N. S., & Nawar, H. A. (2022). Carbon footprint monitoring and mitigation in higher education: a meta-analysis review. Journal of Environmental Sciences (Ain Shams University), 51(11).

[10] Yañez, P., Sinha, A., & Vásquez, M. (2020). Carbon footprint estimation in a university campus: evaluation and insights. Sustainability, 12(1), 181. doi:10.3390/su12010181

[11] Clabeaux, R., Carbajales-Dale, M., Ladner, D., & Walker, T. (2020). Assessing the carbon footprint of a university campus using a life cycle assessment approach. Journal of Cleaner Production, 273, 122600. doi:10.1016/j.jclepro.2020.122600

[12] Filimonau, V., Archer, D., Bellamy, L., Smith, N., & Wintrip, R. (2021). The carbon footprint of a UK University during the COVID-19 lockdown. Science of the Total Environment, 756, 143964. doi:10.1016/j.scitotenv.2020.143964

[13] Mustafa, A., Kazmi, M., Khan, H. R., Qazi, S. A., & Lodi, S. H. (2022). Towards a carbon neutral and sustainable campus: case study of NED University of Engineering and Technology. Sustainability, 14(2), 794. doi:10.3390/su14020794

[14] Cano, N., Berrio, L., Carvajal, E., & Arango, S. (2022). Assessing the carbon footprint of a Colombian university campus using the UNE-ISO 14064–1 and WRI/WBCSD GHG Protocol Corporate Standard. Environmental Science and Pollution Research. doi:10.1007/s11356-022-22119-4

[15] Osorio, A. M., Úsuga, L. F., Vásquez, R. E., Nieto-Londoño, C., Rinaudo, M. E., Martínez, J. A., & Leal Filho, W. (2022). Towards carbon neutrality in higher education institutions: case of two private universities in Colombia. Sustainability, 14(3), 1774. doi:10.3390/su14031774

[16] Battistini, R., Passarini, F., Marrollo, R., Lantieri, C., Simone, A., & Vignali, V. (2023). How to assess the carbon footprint of a large university? The case study of University of Bologna's multicampus organization. Energies, 16(1), 166. doi:10.3390/en16010166

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Thank you

Questions and feedback

